Alexandr Prozorov

Experimental Nuclear and Particle Physicist

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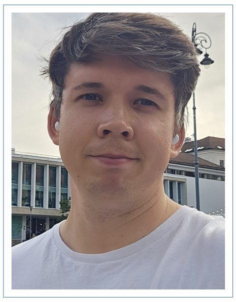

Research profile

I am an experimental nuclear physicist. I work on jets and heavy flavor in heavy-ion collisions, and on electromagnetic and hadronic calorimetry as well as working on software. At the moment that means machine-learning unfolding of jet observables at STAR, and exclusive vector-meson studies with the ePIC backward hadronic calorimeter at Electron-Ion Collider. I am comfortable across the whole analysis chain, from event generators and Geant4 simulation to reconstruction and statistical analysis. I also write AI-assisted software that the collaboration now use as search tool. I am an AI enthusiast and I like teaching: I am convener at the ePIC user-learning Working Group and write much of its training material, and I work with the HEP Software Foundation.

Education

2017–2023 **Ph.D. in Nuclear Physics**, *Charles University*, Prague, Czech Republic
Thesis: *Neutral Meson Production in Ag+Ag Collisions at 1.58 A GeV with HADES Electromagnetic Calorimeter*
Supervisor: RNDr. Andrej Kugler, CSc.

2012–2017 **Ing. (equiv. M.Sc.) in Electronics and Automation of Physical Plants**, *Tomsk Polytechnic University*, Tomsk, Russia, GPA: 5.0/5.0

2015–2016 **Exchange semester**, *Czech Technical University*, Prague, Czech Republic, GPA: 5.0/5.0

Professional experience

2023–present **Researcher**, *Czech Technical University*, Prague, Czech Republic
Jet and heavy-flavor measurements in the STAR experiment at RHIC. Simulation and design studies for the ePIC experiment at the Electron-Ion Collider, work on AI-assistance and documentation tools for both collaborations.

2025 **Research stay**, *Yale University*, New Haven, Connecticut, USA
Jet physics in heavy-ion collisions; ML-based unfolding of jet substructure observables.

2025 **Research stay**, *Brookhaven National Laboratory*, Upton, New York, USA
Work in the BNL Nuclear and Particle Physics Software group on an AI assistant for long-term access to RHIC knowledge (SciBot).

2019–2023 **Researcher / collaboration appointment**, *GSI Helmholtzzentrum für Schwerionenforschung*, Darmstadt, Germany
Detector and analysis work for HADES: ECAL calibration, beam-time operations, detector assembly, and collaboration service.

2017–2023 **Researcher**, *Nuclear Physics Institute, Czech Academy of Sciences*, Řež, Czech Republic
Electromagnetic calorimetry, detector calibration and operations. Analysis of neutral-meson production in heavy-ion collisions inside the HADES collaboration.

2017 **Diploma internship**, *Czech Academy of Sciences*, Prague, Czech Republic
Simulation study of detector-granularity effects on particle-flow observables(directed and elliptic) in relativistic heavy-ion collisions.

2016 **Summer internship**, *National Nuclear Center*, Kurchatov, Kazakhstan
Monte Carlo modeling of a gamma spectrometer with MCNP5.

Research activities

2023–present **Backward hadronic calorimeter: design and physics performance**, *ePIC / EIC*
Position and energy resolution studies, cell-segmentation optimization, from single-particle Geant4 scans. Coherent diffractive $\phi(1020) \to K^+K^-$ production in e+Au collisions. Events come from the Sartre dipole model and go through the full ePIC simulation and reconstruction chain, with a BeAgle background with using Backward Hadronic calorimeter.

2023–present **Jet physics with machine-learning unfolding**, *STAR / RHIC*
D^0-tagged jet measurements unfolded with OmniFold, a machine-learning method that unfolds several observables at once without binning. Also inclusive jet cross-sections in Au+Au and p+p collisions at $\sqrt{s_{NN}} = 200$ GeV, with trigger and detector-response corrections.

2017–2023 **Neutral-meson production with the HADES ECAL**, *HADES / GSI-FAIR*
π^0 and η yields and collective flow in Ag+Ag collisions at 1.58 A GeV ($\sqrt{s_{NN}} = 2.55$ GeV). This was the Ph.D. thesis topic. Responsible for the ECAL calibration and for the efficiency and systematic-uncertainty estimates.

Scientific leadership and service

2026–present **Convener, User Learning Working Group**, *ePIC Collaboration, Software & Computing Coordination Office*
Co-convener (with S. Kay) of the working group that handles onboarding and further software and computing training for ePIC members. Run the ePIC training program of 16 tutorials (eic.github.io/documentation/tutorials).

2025–present **STAR Juniors Representative**, *STAR Collaboration*
Elected by the junior members of the collaboration to represent them.

2023–2026 **Co-organizer**, *Particle Physics Users Workshop: From STAR to EIC*

May 2025 **Organizer**, *STAR Regional Workshop*, Prague, Czech Republic

Jul 2020 **Organizing team**, *ICHEP 2020*, Prague, Czech Republic

Dec 2019 **Local organizer**, *HADES Analysis Meeting*, Řež, Czech Republic

Honors and awards

2025 **Early Career Award**, *STAR Collaboration*
For building and deploying an AI assistant for the collaboration.

Software and open-source contributions

2026–present **Author**, *EIC Smart Search (github.com/eic/eic_smart_search)*
Author of the retrieval-augmented-generation (RAG) service for the search on eic.github.io. It combines keyword and meaning-based search, rewrites the question before searching, and reranks the results afterwards. Access control keeps collaboration-internal material internal. The index covers the EIC websites, code repositories, wiki and document archives.

2026–present **Contributor**, *EIC Software Website (eic/eic.github.io)*
One of the main contributors. Rebuilt the front end, wrote the AI search interface, and reorganized the tutorials and navigation.

2026–present **Author**, *ePIC AI/MCP tutorial and tooling*
Author of *Generative AI for Physics Analysis*, a lesson that teaches AI-assisted analysis through a real $\Lambda^0 \to p\pi^-$ measurement on ePIC reconstructed data. Author of `eic-mcp`, a launcher and service supervisor for the EIC Model Context Protocol servers.

2026 **Contributor**, *EIC software infrastructure*
Small improvement of MCP services, e.g. `xrootd-mcp-server`. Packaging and container work.

2024–present **Author**, *Training material for STAR and EIC workshops*
Self-contained tutorials for collaboration workshops and schools. There is an introductory ROOT course built on CERN Open Data ($J/\psi \to \mu^+\mu^-$), a walkthrough of the STAR analysis environment and a set of Pythia jet exercises. Everything runs in the browser through Binder or GitHub Codespaces, so participants do not have to install anything.

2025–present **Author and maintainer**, *STAR Juniors portal (star-juniors.github.io)*
I built and run the onboarding and documentation site for all STAR members. It covers the computing environment, containers and tape storage, the analysis frameworks, and the collaboration publication procedure.

2026 **Author**, *HEP jobs board*
Open-source tool that collects high-energy and nuclear physics job postings every day and ranks them.

Nov 2024 **Contributor**, *CERN ROOT Hackathon*
Contribution released in ROOT 6.36.

Detector operations and collaboration service

2019–2023 **ECAL calibration**, *HADES / GSI-FAIR*
Calibration of the electromagnetic calorimeter for Ag+Ag collisions at $\sqrt{s_{NN}} = 2.42$ GeV and for pp collisions at $\sqrt{s_{NN}} = 3.6$ GeV.

2019, 2022 **Shift leader and detector operator**, *HADES / GSI-FAIR*
Data-taking shifts as DAQ/ECAL/TOF operator during the Ag+Ag campaign.

2021–2022 **Detector assembly**, *HADES / GSI-FAIR*
Final ECAL sector assembly on site at GSI.

Sep 2020–Jan 2021 **Erasmus+ / GET_INvolved fellow**, *GSI/FAIR*

2019–2020 **Detector installation**, *HADES / GSI-FAIR*
ECAL sector installation and photomultiplier magnetic shielding on site at GSI.

2017–2018 **DAQ and detector support**, *HADES / GSI-FAIR*
FPGA programming for the data-acquisition system. ECAL cabling and TOF maintenance.

Selected presentations

May 2026 **Contributed talk**, *CHEP 2026, 28th International Conference on Computing in High Energy and Nuclear Physics*, Bangkok, Thailand

Mar 2026 **Contributed talk**, *Rencontres de Moriond, QCD and High Energy Interactions*, La Thuile, Italy

Oct 2025 **Contributed talk**, *APS Division of Nuclear Physics Meeting (DNP 2025)*, Chicago, Illinois, USA

Sep 2024 **Poster**, *Hard Probes 2024*, Nagasaki, Japan

Jun 2023 **Contributed talk**, *MESON 2023, 17th International Workshop on Meson Physics*, Krakow, Poland

Jan 2023 **Contributed talk**, *59th International Winter Meeting on Nuclear Physics*, Bormio, Italy

May 2022 **Contributed talk**, *FAIR Next Generation Scientists, 7th Edition*, Paralia, Greece

Apr 2022 **Poster**, *Quark Matter 2022*, Krakow, Poland

Jun 2021 **Contributed talk**, *MESON 2021, 16th International Workshop on Meson Physics*, Krakow, Poland

May 2019 **Contributed talk**, *FAIR Next Generation Scientists, 6th Edition*, Arenzano, Italy

Dec 2018 **Contributed talk**, *Zimányi School and Analytic Hydrodynamics in High Energy Physics*, Budapest, Hungary

Publications

Summary 40 publications indexed on INSPIRE-HEP (STAR, HADES and MPD collaboration papers), 613 citations, h-index 10 (August 2026). Complete lists: ORCID, INSPIRE-HEP, ResearcherID.

Sole-author and small-collaboration publications

2025 M. Atif, V. Garonne, E. Lancon, J. Lauret, **A. Prozorov**, M. Vranovsky, "AI-Powered Assistant for Long-Term Access to RHIC Knowledge," arXiv:2509.09688 [cs.IR] (2025).

2024 **A. Prozorov** (HADES Collaboration), "Neutral meson production in Ag+Ag at $\sqrt{s_{NN}} = 2.55$ GeV," *EPJ Web Conf.* **291**, 04001 (2024).

2023 **A. Prozorov** (HADES Collaboration), "Neutral mesons flow and yields in Ag+Ag at 1.58 AGeV at HADES," *PoS* **FAIRness2022**, 048 (2023).

2023 **A. Prozorov**, "Neutral Meson Production in Ag+Ag Collisions at 1.58 A GeV with HADES Electromagnetic Calorimeter," Ph.D. thesis, Charles University, Prague (2023).

2020 **A. Prozorov**, "Simulation study of effects induced by final granularity of detector in particle flow," *J. Phys. Conf. Ser.* **1667**, 012034 (2020).

2015 **A. A. Prozorov**, A. Yu. Trifonov, A. V. Shapovalov, "Asymptotic behavior of the one-dimensional Fisher–Kolmogorov–Petrovskii–Piskunov equation with anomalous diffusion," *Russ. Phys. J.* **58**, 399–409 (2015).

Selected collaboration publications (co-author)

2026 STAR Collaboration (B. E. Aboona *et al.*), "Measuring spin correlation between quarks during QCD confinement," *Nature* **650**, 65–71 (2026).

2025 STAR Collaboration (B. E. Aboona *et al.*), "Light nuclei femtoscopy and baryon interactions in 3 GeV Au+Au collisions at RHIC," *Phys. Lett. B* **864**, 139412 (2025).

2022 HADES Collaboration (R. Abou Yassine *et al.*), "Measurement of global polarization of Λ hyperons in few-GeV heavy-ion collisions," *Phys. Lett. B* **835**, 137506 (2022).

2019 HADES Collaboration (J. Adamczewski-Musch *et al.*), "Strong absorption of hadrons with hidden and open strangeness in nuclear matter," *Phys. Rev. Lett.* **123**, 022002 (2019).

Technical skills

Programming C++, Python, Bash, SQL, LaTeX, JavaScript

HEP software ROOT, Geant4, DD4hep, JANA2, PODIO/EDM4hep, HepMC3, uproot, XRootD, Rucio

Event generators Sartre (dipole/saturation model), Pythia8, MCNP5

Computing Linux, Docker/Singularity, Spack, Git and GitHub/GitLab CI/CD, batch systems (HTCondor, Slurm), PostgreSQL, Jupyter

AI and LLM tooling Retrieval-augmented generation, vector databases, embedding models, Model Context Protocol (MCP) servers, agentic coding workflows, TensorFlow

Instrumentation Electromagnetic and hadronic calorimetry, detector calibration and commissioning, DAQ operations, FPGA programming

Languages English (fluent), Russian (native), Czech (working knowledge)